Exp. No: 06

12.8.26

Hamming Code

Aim:

Write a program to implement error detection and correction using hamming code concept. Make test run to input data stream and verify correction feature.

Code:

```
def calc_redundant_bits(m):
    for r in range(m+1):
        if (2**r) >= (m+r+1):
            return r

def main():
    data = input().strip()
    m = len(data)
    r = calc_redundant_bits(m)
    n = m+r
    print(f"Number of r needed: {r}")
    print(f"Total frame length: {n}")
    code = ['0'] * (n+1)
    j = 0
    for i in range(1, n+1):
        if (i & (i-1)) != 0:
            code[i] = data[-(j+1)]
            j += 1
    for i in range(r):
        parity_pos = 2**i
        val = 0
        for pos in range(1, n+1):
            if (pos & parity_pos) != 0 and
                pos != parity_pos:
                if code[pos] == '1':
                    val = 1
```

```
        code[parity_pos] = str(val)
    transmitted_str = "".join(code[1:][::-1])

    print(f"Data to be transmitted: {transmitted_str}")

change = input("\nTransmitted bit change? (Yes/No): ").strip().lower()

if change == "yes":
    received_str = input("Corrupt data: ").strip()

    if len(received_str) != n:
        print(f"Error: received data must be exactly {n} bits")
    else:
        received_code = list(received_str)

        error_pos = 0
        for i in range(r):
            parity_pos = 2 ** i
            var = 0
            for pos in range(1, n+1):
                if (pos & parity_pos) != 0:
                    if received_code[pos] == '1':
                        var ^= 1
            if var != 0:
                error_pos += parity_pos

        if error_pos == 0:
            print("No error detected")
        else:
            print(f"Error detected at position: {error_pos}")
            received_code[error_pos] = '0' if received_code[error_pos] == '1' else '1'
            corrected = "".join(received_code[1:][::-1])
            print(f"Corrected data: {corrected}")

if __name__ == "__main__":
    main()
```

Output:

1. Enter data bits: 101100
   Number of redundant bits: 4
   Total frame length: 10
   Data to be transmitted: 1011100011
   Transmitted bit change? (Yes/No): Yes
   Enter corrupt data: 1011101011
   Error detected
   Wrong bit at position: 4
   Corrected data stream: 1011100011

2. Enter data bits: 1011
   Number of redundant bits: 3
   Total data frame length: 7
   Data to be Transmitted: 0110011
   Transmitted bit change? (Yes/No): No
   No error detected.

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Result:

Thus, the hamming code was implemented successfully for error detection and correction